CMPG 322 - Decision Support Systems (DSS) 2
SU1 - Decision Theory 1



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Outline

1. Overview
2. Quantitative analysis and Decision Support Systems
3. Steps in decision making
4. Types of decision making environments
5. Decision making under uncertainty
6. Decision making under risk
7. Summery
8. Applied Assignment A
9. Questions

Overview

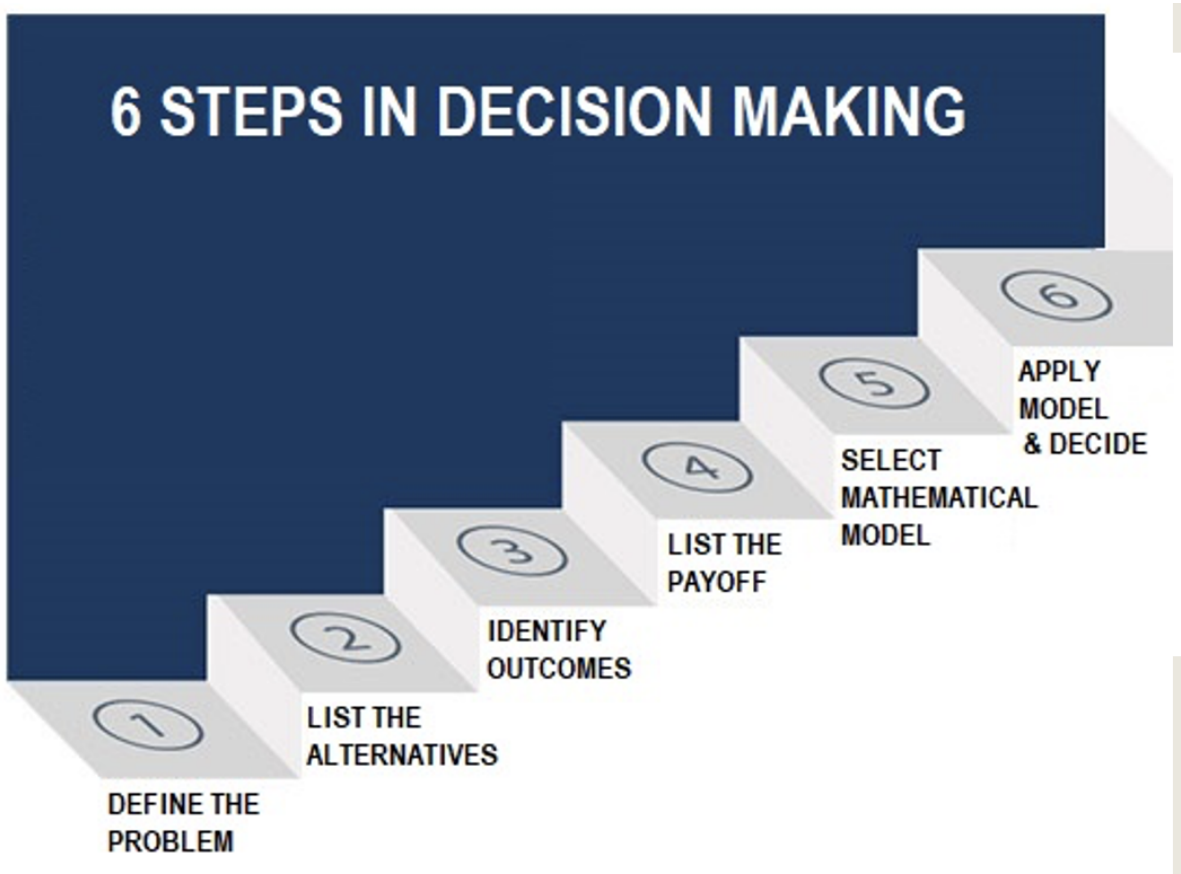
- Quantitative analysis vs. decision support systems
- Six steps in decision making
- Types of decision making environments

(This study unit is based on sections 3.1 – 3.5 of Chapter 3 in the textbook)

- Quantitative analysis is the use of mathematical and statistical methods in finance
- A decision support system is an information system that supports business or organizational decision-making activities

Quantitative analysis is the scientific approach to managerial decision making.

Steps in decision making



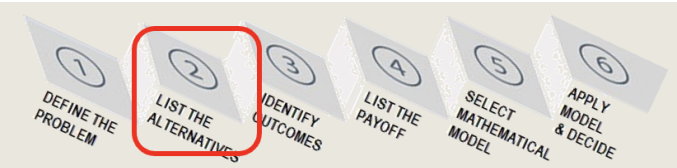
Example:
John Thompson is the founder and president of Thompson Lumber Company, a profitable firm located in Portland, Oregon. The problem that John identifies is whether to expand his product line by manufacturing and marketing a new product, backyard storage sheds.



TABLE 3.1 Decision Table with Conditional Values for Thompson Lumber

	STATE OF NATURE	
	FAVORABLE MARKET	UNFAVORABLE MARKET
ALTERNATIVE	($)	($)
Construct a large plant	200,000	−180,000
Construct a small plant	100,000	−20,000
Do nothing	0	0

Note: It is important to include all alternatives, including "do nothing."



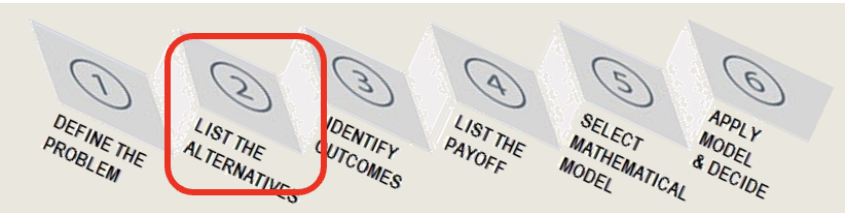
Steps in decision making

TABLE 3.1 Decision Table with Conditional Values for Thompson Lumber

Those outcomes over which the decision maker has little or no control are called **states of nature**.

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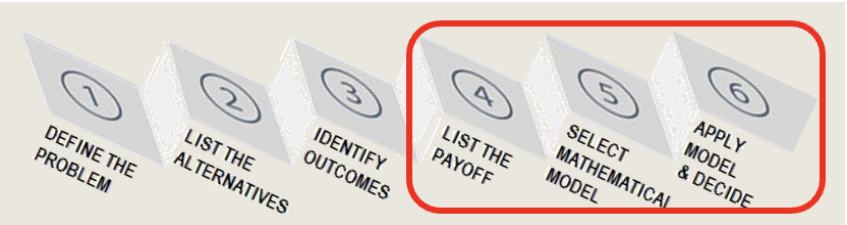


Steps in decision making

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Types of decision making environments

- Decision making under certainty
  - The decision maker knows with certainty the consequences of every alternative or decision choice
- Decision making under uncertainty
  - The decision maker does not know the probabilities of the various outcomes
- Decision making under risk
  - The decision maker knows the probabilities of the various outcomes

Decision making under uncertainty

Criteria:

1. *Maximax (optimistic)*
2. *Maximin (pessimistic)*
3. *Criterion of realism (Hurwicz)*
4. *Equally likely (Laplace)*
5. *Minimax regret*

Can be computed directly from the decision (payoff) table

Requires use of an opportunity loss table

Calculation of:

1. Expected monetary value (EMV)
2. Expected Value of Perfect Information (EVPI)
3. Expected Opportunity Loss (EOL)
4. Sensitivity Analysis

Summery:

- Quantitative analysis vs. decision support systems.
- Six steps in decision making.
- Types of decision making environments.

Design a virtual business that:

- Manufactures two or more products;
- Performs two or more services;
- Has a venue from which products can be sold/services provided;
- Has a staff of more than 10 employees; and
- Has two or more competitive businesses in the market.

Questions

